

1.

(a) incorrect

If $c = 1$, the right side becomes a constant. Since $k \geq 1$, n^k grows to infinity and cannot be bounded by a constant.

(b) correct

$n! = n \times (n - 1) \times \dots \times 1$, which is strictly less than n^n for large n .

Therefore, $\lg(n!) \leq \lg(n^n) = n \lg(n)$.

Since $n \lg(n)$ grows much slower than n^n , $\lg(n!) = O(n^n)$.

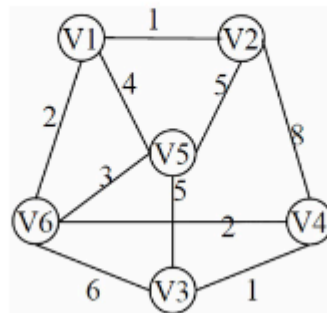
(c) correct

The dominant term is $n^3 \lg(n)$, it follows $n^3 \lg(n) < n^3 \times n = n^4$ for large n .

Since the dominant term is bounded by n^4 , the entire expression is $O(n^4)$.

2.

2. (30%) Solve the TSP problem in the following figure, starting and ending at V1, with different approaches.



(a) Use the branch and bound algorithm, and show your work step by step.

(b) Give an ILP formulation, and show the detail.

(c) Apply the simulated-annealing (SA) algorithm, and show the detail.

Solution

(a) Use the branch and bound algorithm:

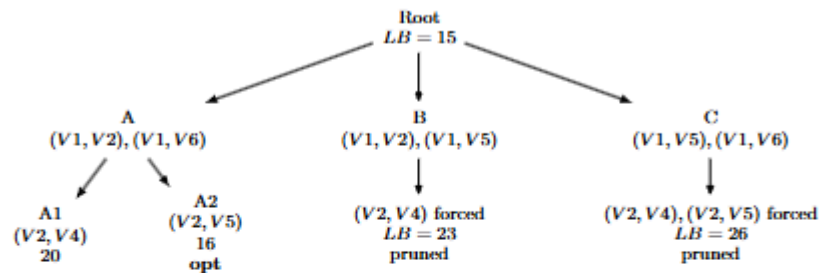
Initial lower bound:

$$LB_0 = \left\lceil \frac{(1+2) + (1+5) + (1+5) + (1+2) + (3+4) + (2+2)}{2} \right\rceil = \left\lceil \frac{29}{2} \right\rceil = 15$$

Branch on V1:

$(V1, V2), (V1, V6), (V1, V2), (V1, V5), (V1, V5), (V1, V6)$

Branch-and-bound tree



For Branch A:

$$(V1, V2), (V1, V6), (V2, V4) \Rightarrow 1 + 2 + 8 + 1 + 5 + 3 = 20$$

$$(V1, V2), (V1, V6), (V2, V5) \Rightarrow 1 + 2 + 5 + 5 + 1 + 2 = 16$$

For Branch B, choosing $(V2, V5)$ forms the subtour

$$V1 \rightarrow V2 \rightarrow V5 \rightarrow V1$$

so $(V2, V4)$ is forced, and

$$LB_B = 1 + 4 + 8 + 1 + 6 + 3 = 23 > 16$$

Hence Branch B is pruned.

For Branch C, since $V2$ cannot connect to $V1$, both $(V2, V4)$ and $(V2, V5)$ are forced, and

$$LB_C = 4 + 2 + 8 + 5 + 1 + 6 = 26 > 16$$

Hence Branch C is pruned.

Therefore, the optimal tour is

$$\boxed{V1 \rightarrow V2 \rightarrow V5 \rightarrow V3 \rightarrow V4 \rightarrow V6 \rightarrow V1}$$

with total cost

$$\boxed{16}$$

(b) Give an ILP formulation:

Define a binary variable for each existing edge:

$$x_{ij} = \begin{cases} 1, & \text{if edge } (i, j) \text{ is selected} \\ 0, & \text{otherwise} \end{cases}$$

Objective:

$$\min z = 1x_{12} + 4x_{15} + 2x_{16} + 8x_{24} + 5x_{25} + 1x_{34} + 5x_{35} + 6x_{36} + 2x_{46} + 3x_{56}$$

Degree constraints:

$$x_{12} + x_{15} + x_{16} = 2$$

$$x_{12} + x_{24} + x_{25} = 2$$

$$x_{34} + x_{35} + x_{36} = 2$$

$$x_{24} + x_{34} + x_{46} = 2$$

$$x_{15} + x_{25} + x_{35} + x_{56} = 2$$

$$x_{16} + x_{36} + x_{46} + x_{56} = 2$$

Subtour elimination:

$$\sum_{\substack{i \in S, j \notin S \\ (i,j) \in E}} x_{ij} \geq 2, \quad \forall S \subset \{1, 2, 3, 4, 5, 6\}, 1 \leq |S| \leq 5$$

Binary constraints:

$$x_{ij} \in \{0, 1\}, \quad \forall (i, j) \in E$$

One optimal solution is

$$x_{12} = x_{25} = x_{35} = x_{34} = x_{46} = x_{16} = 1$$

and all other variables are 0.

Thus the optimal tour is

$$\boxed{V1 \rightarrow V2 \rightarrow V5 \rightarrow V3 \rightarrow V4 \rightarrow V6 \rightarrow V1}$$

with

$$\boxed{z = 16}$$

(c) Apply the simulated-annealing (SA) algorithm:

Represent a solution as an ordering starting and ending at $V1$. For example,

$$[V1, V2, V4, V6, V3, V5, V1]$$

Use swap of two internal vertices as the neighbor move. Accept a new solution with

$$\Delta = f(\text{new}) - f(\text{current})$$

If $\Delta \leq 0$, accept it directly; otherwise accept it with probability

$$P = e^{-\Delta/T}$$

Use:

$$T_0 = 10, \quad \alpha = 0.8, \quad T \leftarrow \alpha T$$

Start with

$$\begin{aligned} S_0 &= [V1, V2, V4, V6, V3, V5, V1] \\ f(S_0) &= 1 + 8 + 2 + 6 + 5 + 4 = 26 \end{aligned}$$

Swap V6 and V3:

$$\begin{aligned} S_1 &= [V1, V2, V4, V3, V6, V5, V1] \\ f(S_1) &= 1 + 8 + 1 + 6 + 3 + 4 = 23 \\ \Delta &= 23 - 26 = -3 < 0 \end{aligned}$$

Accept.

Swap V6 and V5:

$$\begin{aligned} S_2 &= [V1, V2, V4, V3, V5, V6, V1] \\ f(S_2) &= 1 + 8 + 1 + 5 + 3 + 2 = 20 \\ \Delta &= 20 - 23 = -3 < 0 \end{aligned}$$

Accept.

Swap V4 and V5:

$$\begin{aligned} S_3 &= [V1, V2, V5, V3, V4, V6, V1] \\ f(S_3) &= 1 + 5 + 5 + 1 + 2 + 2 = 16 \\ \Delta &= 16 - 20 = -4 < 0 \end{aligned}$$

Accept.

Therefore, the final best solution is

$$\boxed{V1 \rightarrow V2 \rightarrow V5 \rightarrow V3 \rightarrow V4 \rightarrow V6 \rightarrow V1}$$

with cost

$$\boxed{16}$$

Grading Rubric

(a) Branch and Bound: 10 points

Item	Points	Criteria
Branch-and-bound tree	2	Draws a branch/search tree. It does not need to be exactly the same as the reference solution.
Correct branch cost / bound calculation	3	Shows cumulative costs, current best value, bounds, or reasonable comparison of branch costs.
Pruning	2	Uses pruning, such as X , exceeding the bound, infeasible paths, repeated vertices, or paths that cannot return to $V1$.
Correct answer	3	Finds the correct optimal tour, or its reverse, with total cost 16.
Total	10	

The correct optimal tour is

$$V1 \rightarrow V2 \rightarrow V5 \rightarrow V3 \rightarrow V4 \rightarrow V6 \rightarrow V1$$

or equivalently its reverse:

$$V1 \rightarrow V6 \rightarrow V4 \rightarrow V3 \rightarrow V5 \rightarrow V2 \rightarrow V1.$$

The optimal cost is

$$\boxed{16}.$$

(b) ILP Formulation: 10 points

Item	Points	Criteria
Binary decision variables	2	Defines binary variables such as x_{ij} to indicate whether edge (i, j) is selected.
Objective function	2	Correctly minimizes total edge cost, e.g. $\min \sum_{(i,j) \in E} w_{ij} x_{ij}.$
Degree constraints	2	Ensures each vertex has degree 2, or in directed form, in-degree 1 and out-degree 1.
Subtour elimination constraints	3	Includes valid subtour elimination constraints, such as cut constraints or MTZ constraints.
Binary constraints	1	States $x_{ij} \in \{0, 1\}.$
Total	10	

Note that the constraint

$$\sum x_{ij} = 6$$

only states that six edges are selected. It does not replace the degree constraints or subtour elimination constraints.

(c) Simulated Annealing: 10 points

Item	Points	Criteria
SA basic setup	3	Includes solution representation, neighbor move, cost function, and acceptance rule.
Temperature schedule / stopping rule	3	Includes a cooling rule, e.g. $T \leftarrow \alpha T$, and a stopping rule, such as frozen condition, maximum iterations, or time limit.
Actual SA iterations	4	Applies SA to this problem by listing actual solutions such as S_0, S_1 , their costs, Δ , and accept/reject decisions. Ideally obtains or approaches cost 16.
Total	10	

The SA acceptance rule should be:

$$\Delta = f(S') - f(S),$$

$$\Delta \leq 0 \Rightarrow \text{accept } S',$$

$$\Delta > 0 \Rightarrow \text{accept } S' \text{ with probability } e^{-\Delta/T}.$$

Acceptable neighbor moves include:

swap two internal vertices,

2-opt moves, or other random perturbations, provided that feasibility is checked.

3.

Solution

Since the problem statement asks for the gain of **net cut**, the black dot can be interpreted as a branching point of the same net. Thus, the nets are:

$$n_1 = \{C_1, C_2\}, \quad n_2 = \{C_1, C_3, C_4\}, \quad n_3 = \{C_2, C_4\}$$

The initial partition is:

$$A = \{C_1, C_2\}, \quad B = \{C_3, C_4\}$$

Before moving C_1 and C_4 , the cut nets are:

$$n_2, n_3$$

Thus,

$$\text{cut}_{\text{old}} = 2$$

After simultaneously moving C_1 and C_4 to their opposite sides, the partition becomes:

$$A = \{C_2, C_4\}, \quad B = \{C_1, C_3\}$$

After the move, the cut nets are:

$$n_1, n_2$$

Thus,

$$\text{cut}_{\text{new}} = 2$$

Therefore, the gain is:

$$g = \text{cut}_{\text{old}} - \text{cut}_{\text{new}} = 2 - 2 = 0$$

Hence, under the **net-cut** definition stated in the problem,

$$\boxed{g = 0}$$

4.

Solution

The correct statement is:

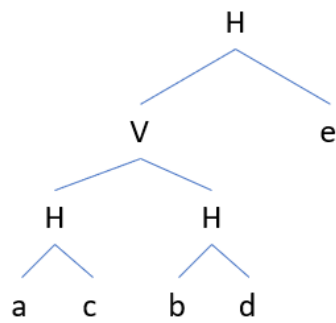
$$\boxed{(c) \text{ only}}$$

(a) **Incorrect.** A general slicing floorplan does not have a one-to-one correspondence with a Polish expression. The same slicing floorplan may be represented by multiple Polish expressions. A unique representation usually requires additional restrictions, such as using a normalized Polish expression.

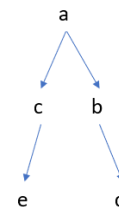
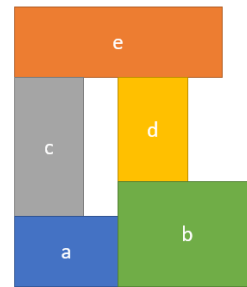
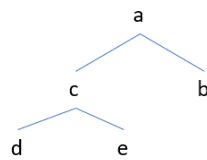
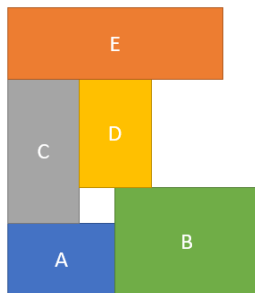
(b) **Incorrect.** A general slicing floorplan does not have a one-to-one correspondence with a slicing tree. The same slicing floorplan may have different slicing-tree representations, especially when consecutive cuts have the same direction. A more unique representation requires restrictions such as using a skewed slicing tree.

(c) **Correct.** A sequence pair determines the horizontal and vertical relative relations between modules. These relations correspond to the horizontal and vertical constraint graphs. Since a TCG representation is also composed of these two types of constraint graphs, under the standard definition, a TCG representation and a sequence-pair representation can be regarded as having a one-to-one correspondence.

5.



- a) [1pt for 1 mistake]
 b) $ac=(3,6) \rightarrow bd=(4,6) \rightarrow abcd=(7,6) \rightarrow abcde=(7,8)$, $8*7=56$ [2pts for process, 3pts for correct]
 c) pic, $S+\{ecdab, ecadb\}$ $S-\{acbde\}$ [2pts for pic, 3pts for correct]
 d) HCG, VCG $8*7=56$ [3pts for process, 2pts for correct]
 e) compaction [2pts for pic, 3pts for tree]



- f) $8*7=56$ [3pts for process or pic, 2pts for answer]
 g) TCG*2 [1pt for 1 mistake]

